

PROJECT REPORT: SMART HIRE

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Abstract

The rapid expansion of digital job applications has introduced significant bottlenecks in modern talent acquisition. **Smart Hire** is an intelligent talent acquisition and assessment platform designed to streamline hiring workflows through automation. Developed utilizing the Antigravity agentic framework, the platform shifts recruitment from manual, keyword-dependent resume screening to an autonomous, context-aware evaluation process. By leveraging Natural Language Processing (NLP) and multi-agent coordination, Smart Hire extracts candidate information, generates compatibility scores against specific job descriptions, and provides actionable insights for both recruiters and job seekers.

Introduction

Recruitment systems serve as the critical bridge between talented individuals and organizational needs. However, as the volume of applications scales, traditional Applicant Tracking Systems (ATS) struggle to accurately evaluate candidate potential, often filtering out qualified individuals due to minor formatting issues or missing exact-match keywords. Smart Hire introduces a paradigm shift by integrating advanced AI methodologies into the recruitment pipeline. By employing the Antigravity environment to coordinate intelligent agents, the system automates end-to-end resume analysis, candidate ranking, and communication, transforming a traditionally time-consuming administrative task into a highly optimized, data-driven workflow.

Problem Statement

Manual eligibility checking and resume screening require recruiters to individually verify academic details, technical skills, and experience against specific company criteria. This process is highly inefficient, prone to human error, and susceptible to unconscious bias. Furthermore, candidates often lack visibility into why they were rejected or how their skills align with industry expectations. There is a critical need for a centralized, intelligent platform that can autonomously parse unstructured resume data, objectively evaluate candidate suitability, and reduce the manual overhead of the hiring lifecycle.

Research Gap

While modern recruitment platforms incorporate basic machine learning for keyword matching, they largely operate as passive databases rather than active decision-making engines. The primary research gap lies in the lack of **agentic workflow automation** within recruitment tools. Existing systems fail to employ autonomous agents capable of handling complex, multi-step tasks—such as dynamically adjusting evaluation criteria, performing context-aware skill gap analyses, and autonomously scheduling or communicating with candidates—without constant human intervention.

Literature Review & Related Work

The evolution of recruitment technology has transitioned through three distinct phases:

- **Manual Processing:** Heavy reliance on spreadsheets and manual verification, resulting in slow turnaround times.
- **Traditional ATS:** Introduction of digital databases and boolean keyword filtering. While improving storage, these systems suffer from high false-positive and false-negative rates in candidate evaluation.
- **AI-Assisted Matching:** Recent studies propose the use of machine learning algorithms to compare candidate profiles against job descriptions.

Smart Hire builds upon this third phase by introducing the Antigravity framework, which moves beyond simple matching to deploy coordinated AI agents that manage discrete aspects of the pipeline (e.g., parsing, scoring, and workflow management) simultaneously.

Methodology

The development and operation of Smart Hire rely on a multi-agent architectural approach:

- **Agent-Assisted Development:** The Antigravity environment is utilized to structure the software, handle complex API integrations, and manage the underlying logic of the platform.
- **Data Extraction (NLP):** Natural Language Processing techniques are applied to unstructured resumes to accurately extract entities such as education, technical proficiencies, and work experience.
- **Evaluation Algorithms:** The system calculates a dynamic ATS compatibility score by weighing extracted skills against the semantic requirements of the job description, rather than relying on strict string-matching.
- **Automated Workflows:** Dedicated agents handle post-evaluation tasks, such as generating personalized feedback for candidates and compiling ranked shortlists for recruiters.

System Overview

The Smart Hire platform is divided into three core functional modules:

Module	Primary Functions
Job Seeker Dashboard	Profile creation, resume upload, real-time ATS score viewing, and automated skill-gap feedback.
Recruiter Dashboard	Job requirement posting, automated candidate shortlisting, and pipeline analytics.
Agent Coordinator (Backend)	Manages the Antigravity agents responsible for NLP parsing, database management, and autonomous communication routing.

Smart Hire Architecture Diagram Smart Hire Job Seeker Platform Workflow Diagram

Figures 1 & 2: Smart Hire Architecture and Job Seeker Workflow Diagrams

Result

The implementation of Smart Hire significantly accelerates the talent acquisition process. The intelligent automation of resume screening reduces the manual workload on recruitment staff while ensuring a more objective, skill-based evaluation of candidates. Recruiters benefit from an organized, centralized pipeline of high-quality matches, and candidates receive a more transparent application experience with actionable insights into their professional profiles.